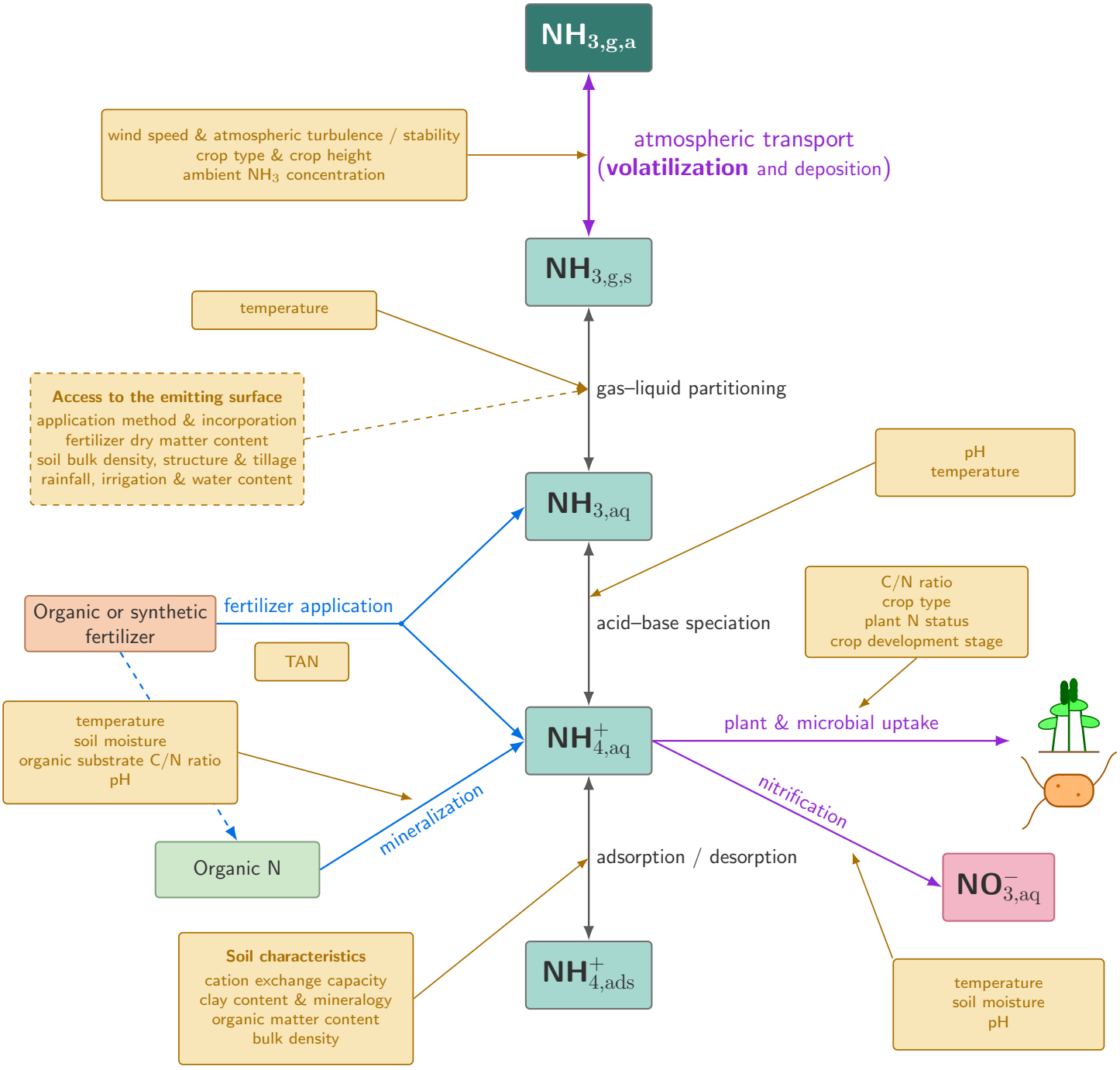

$NH_{3,g,a}$
wind speed & atmospheric turbulence / stability
crop type & crop height
ambient NH_3 concentration
atmospheric transport
(**volatilization** and deposition)
$NH_{3,g,s}$
temperature
gas–liquid partitioning
Access to the emitting surface
application method & incorporation
fertilizer dry matter content
soil bulk density, structure & tillage
rainfall, irrigation & water content
pH
temperature
$NH_{3,aq}$
Organic or synthetic fertilizer
fertilizer application
TAN
acid–base speciation
C/N ratio
crop type
plant N status
crop development stage
temperature
soil moisture
organic substrate C/N ratio
pH
$NH_{4,aq}^{+}$
plant & microbial uptake
mineralization
nitrification
Organic N
adsorption / desorption
$NO_{3,aq}^{-}$
Soil characteristics
cation exchange capacity
clay content & mineralogy
organic matter content
bulk density
$NH_{4,ads}^{+}$
temperature
soil moisture
pH